AS

ARI SWERDLOW

PRODUCT ENGINEER / COMPUTER VISION AND BROWSER GAMES

I build apps where a phone camera is the controller. Bad lighting, lost tracking, slow phones.

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PROFILE

Independent product engineer. I take messy camera input and turn it into something a normal person can use without reading instructions.

CORE STRENGTHS

- Computer vision product UX
- Browser AR and camera pipelines
- Finite-state game systems
- Simulation and recorded replay
- On-device performance budgets
- Accessibility and recovery design
- Release gates and observability

TECHNOLOGY

TypeScript, React, Next.js, Three.js, React Three Fiber, MediaPipe, TensorFlow.js, WebGL, Python, FastAPI, Playwright, deterministic simulation

WORKING PRINCIPLES

Uncertainty stays visible
Low-confidence input pauses or downgrades a judgment.

Recovery is product state
Permission, tracking loss, and interruption get explicit paths.

Evidence has boundaries
Engineering checks are never relabeled as user outcomes.

CURRENT FOCUS

Available for product engineering and applied computer vision work where software must understand the physical world.

SELECTED PRODUCT WORK / 2025-2026

Jiku Fitness

PRODUCT + ENGINEERING

Built a browser workout for boxing, push-ups and pull-ups that handles lost tracking and rest without freezing.

20 BOXING TESTS / REAL-PHONE TESTING PENDING

AR Baseball

GAME + POSE SYSTEMS

Built a three-inning baseball game controlled by your swing, with fielding, runners and a rival team.

250,000 SIMULATED GAMES / 21 TESTS

Bio-Tactical Edge

RESEARCH + PRODUCT ARCHITECTURE

Designed a tool that turns match video into court positions, player paths, a 3D replay and searchable clips.

77 TESTS

ADDITIONAL SYSTEMS SHIPPED

Jiku Tennis	body-tracked rallies and five-rival progression	33 TESTS
Form: Pickleball	pose, ball, and audio coaching workflows	695 TESTS
ASL Hero	self-hosted hand tracking and learned fallback	DECISION-PATH TESTS
AR Slicer	independent camera layer and thermal-aware quality	RECOVERY GATE
Orbitap	seeded courses and shareable ghost runs	PWA BUILD

ENGINEERING VALIDATION SNAPSHOT

250k	695	5
BASEBALL SIMULATIONS	COACH TESTS	CAMERA PRODUCTS

WHAT I CAN DO FOR YOUR TEAM

Prototype the hard input
Get gestures, pose or video working as a real control.

Make it trustworthy
Plan for bad lighting, lost tracking and privacy from day one.

Ship with evidence
Tests, replays and numbers that say what they don't cover.

AVAILABLE NOW

Product engineering / Applied CV / Interaction systems

These are engineering numbers, not user results. More at ari-swerdlow.vercel.app/evidence

AVAILABLE FOR PRODUCT ENGINEERING + APPLIED CV